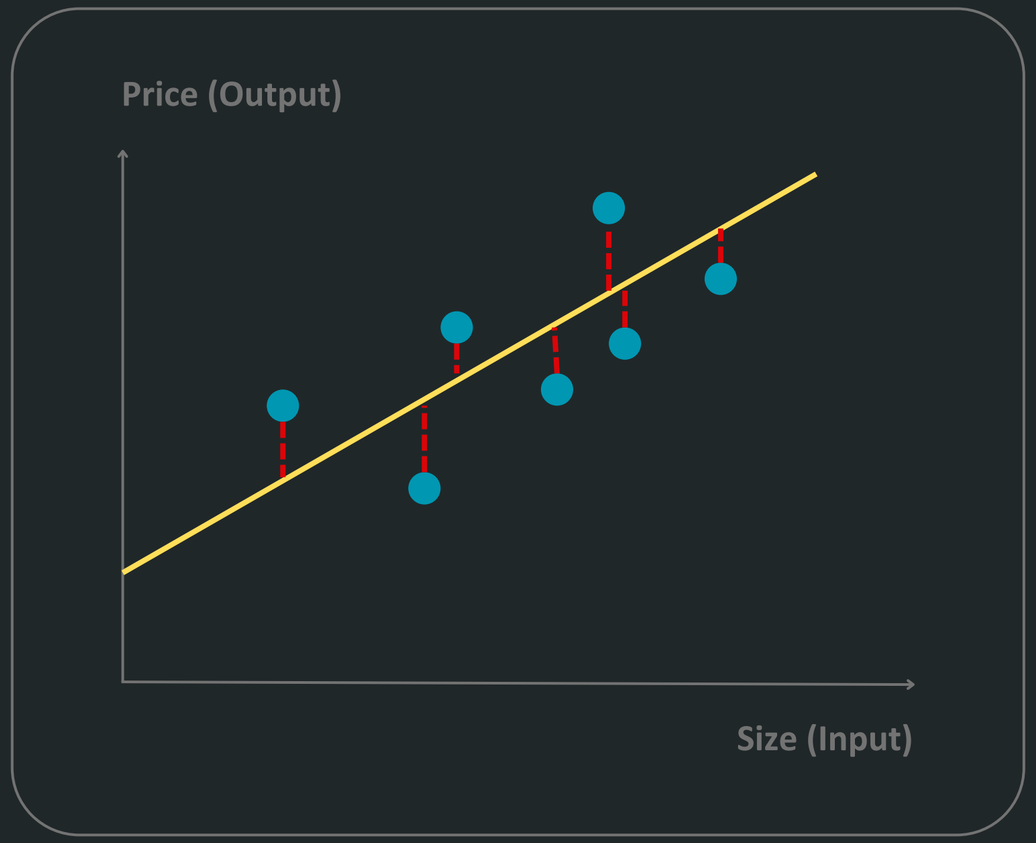
Section 6. Math Foundation for ML

6.1. Why Math matters for ML

How does an ML model learn?



It’s simply finding the line of best fit

Data Point

A single example the model learns from. It shows the true input and the true output

The Model (Line)

The line the model draws to make predictions. It tries to capture the pattern in the data

Error (Gap)

How far the prediction is from the real value. The model reduces this gap to learn better

The Four Pillars

Linear Algebra

Data Structure

Vectors and matrices for data. Computations inside every model.

Statistics

Analysis

Understand data patterns. Measure model performance.

Calculus

Optimization

Gradients for learning. Optimize model weights.

Probability

Uncertainity

Handle uncertainty. Power predictions and classifiers.